

React HOL Solutions -- Part 1

Covers Labs 1--4

Lab 1 -- My First React App

Objective

Create a React app **myfirstreact** and display:

```
Welcome to the first session of React
```

Commands

```
npx create-react-app myfirstreact  
cd myfirstreact  
npm start
```

File Structure

```
src/  
├─ App.js
```

App.js

```
function App() {  
  return (  
    <div>  
      <h1>Welcome to the first session of React</h1>  
    </div>  
  );  
}  
  
export default App;
```

Lab 2 -- Student Management Portal

Objective

Create three components: - Home - About - Contact

Commands

```
npx create-react-app StudentApp
cd StudentApp
```

Structure

```
src/
├── Components/
│   ├── Home.js
│   ├── About.js
│   └── Contact.js
└── App.js
```

Home.js

```
export default function Home() {
  return <h2>Welcome to the Home page of Student Management Portal</h2>;
}
```

About.js

```
export default function About() {
  return <h2>Welcome to the About page of Student Management Portal</h2>;
}
```

Contact.js

```
export default function Contact() {
  return <h2>Welcome to the Contact page of Student Management Portal</h2>;
}
```

App.js

```
import Home from "./Components/Home";
import About from "./Components/About";
```

```
import Contact from "../Components/Contact";

function App() {
  return (
    <>
      <Home />
      <About />
      <Contact />
    </>
  );
}

export default App;
```

Lab 3 -- Score Calculator (Functional Component)

Objective

Accept Name, School, Total and Goal and display percentage.

Structure

```
src/
├── Components/
│   └── CalculateScore.js
├── Stylesheets/
│   └── mystyle.css
└── App.js
```

CalculateScore.js

```
import "../Stylesheets/mystyle.css";

const percent = (total, goal) => ((total / goal) * 100).toFixed(2);

export default function CalculateScore({ name, school, total, goal }) {
  return (
    <div className="card">
      <h2>Student Details</h2>
      <p>Name: {name}</p>
      <p>School: {school}</p>
      <p>Total: {total}</p>
      <p>Score: {percent(total, goal)}%</p>
    </div>
  );
}
```

mystyle.css

```
.card{
border:1px solid #000;
padding:20px;
width:300px;
}
```

App.js

```
import CalculateScore from "../Components/CalculateScore";

function App() {
  return (
    <CalculateScore
      name="Varun"
      school="LNCT"
      total={430}
      goal={500}
    />
  );
}

export default App;
```

Lab 4 -- Lifecycle Methods

Objective

Fetch posts using componentDidMount()

Structure

```
src/  
├── Post.js  
├── Posts.js  
└── App.js
```

Post.js

```
class Post {  
  constructor(id,title,body){  
    this.id=id;  
    this.title=title;  
    this.body=body;  
  }  
}  
  
export default Post;
```

Posts.js

```
import React,{Component} from "react";  
  
class Posts extends Component{  
  
  state={posts:[]};  
  
  async componentDidMount(){  
    const res=await fetch("https://jsonplaceholder.typicode.com/posts");  
    const data=await res.json();  
    this.setState({posts:data});  
  }  
  
  render(){  
    return(  
      <div>  
        {this.state.posts.map(post=>(  
          <div key={post.id}>  
            <h3>{post.title}</h3>  
            <p>{post.body}</p>  
          </div>  
        ))}  
      </div>  
    );  
  }  
}
```

```
}  
  
export default Posts;
```

App.js

```
import Posts from "./Posts";  
  
function App(){  
  return <Posts />;  
}  
  
export default App;
```

React HOL Solutions - Part 2

Covers Labs 5–8

Lab 5 - Styling React Components

Objective

Style the Cohort Tracker application using CSS Modules.

Commands

```
npm install  
npm start
```

Folder Structure

```
src/  
├── Components/  
│   ├── CohortDetails.js  
│   └── CohortDetails.module.css  
└── App.js
```

CohortDetails.module.css

```
.box{
  width:300px;
  display:inline-block;
  margin:10px;
  padding:10px 20px;
  border:1px solid black;
  border-radius:10px;
}

dt{
  font-weight:500;
}
```

CohortDetails.js

```
import styles from "./CohortDetails.module.css";

function CohortDetails({cohort}){

  return(
    <div className={styles.box}>

      <h3
        style={{
          color:cohort.status==="ongoing" ? "green":"blue"
        }}
      >
        {cohort.code}
      </h3>

      <dl>
        <dt>Trainer</dt>
        <dd>{cohort.trainer}</dd>

        <dt>Strength</dt>
        <dd>{cohort.strength}</dd>

        <dt>Status</dt>
        <dd>{cohort.status}</dd>
      </dl>

    </div>
  );
}

export default CohortDetails;
```

App.js

```
import CohortDetails from "../Components/CohortDetails";

const cohort={
  code:"INTADMDF10",
  trainer:"John",
  strength:30,
  status:"ongoing"
};

function App(){
  return <CohortDetails cohort={cohort}/>;
}

export default App;
```

Lab 6 - React Router

Objective

Create routing between Home, Trainers List and Trainer Details.

Install

```
npm install react-router-dom
```

Folder Structure

```
src/
├── trainer.js
├── TrainersMock.js
├── Home.js
├── TrainersList.js
├── TrainerDetails.js
└── App.js
```

trainer.js

```
class Trainer{

  constructor(id,name,email,phone,technology,skills){
    this.id=id;
    this.name=name;
```



```
        this.email=email;
        this.phone=phone;
        this.technology=technology;
        this.skills=skills;
    }

}

export default Trainer;
```

TrainersMock.js

```
import Trainer from "./trainer";

const trainers=[

    new Trainer(
        1,
        "Varun",
        "varun@gmail.com",
        9876543210,
        "React",
        ["React", "Node", "MongoDB"]
    ),

    new Trainer(
        2,
        "Rahul",
        "rahul@gmail.com",
        9876500000,
        "Java",
        ["Spring", "Hibernate"]
    )

];

export default trainers;
```

Home.js

```
function Home(){

    return(
        <h2>Welcome to My Academy Trainers Page</h2>
    );

}
```

```
export default Home;
```

TrainersList.js

```
import trainers from "./TrainersMock";
import {Link} from "react-router-dom";

function TrainersList(){

  return(

    <ul>

      {trainers.map(trainer=>(

        <li key={trainer.id}>

          <Link to={`/trainer/${trainer.id}`}>
            {trainer.name}
          </Link>

        </li>

      ))}

    </ul>

  );

}

export default TrainersList;
```

TrainerDetails.js

```
import {useParams} from "react-router-dom";
import trainers from "./TrainersMock";

function TrainerDetails(){

  const {id}=useParams();

  const trainer=trainers.find(
    t=>t.id===parseInt(id)
  );

  return(
```

```
        <div>

            <h2>{trainer.name}</h2>

            <p>{trainer.email}</p>

            <p>{trainer.phone}</p>

            <p>{trainer.technology}</p>

            <ul>

                {trainer.skills.map(skill=>(

                    <li key={skill}>{skill}</li>

                ))}

            </ul>

        </div>

    );

}

export default TrainerDetails;
```

App.js

```
import {
  BrowserRouter,
  Routes,
  Route,
  Link
} from "react-router-dom";

import Home from "./Home";
import TrainersList from "./TrainersList";
import TrainerDetails from "./TrainerDetails";

function App(){

    return(

        <BrowserRouter>

            <h1>My Academy Trainers App</h1>

            <Link to="/">Home</Link>
```

```
    {" | "}

    <Link to="/trainers">
      Show Trainers
    </Link>

    <Routes>

      <Route
        path="/"
        element={<Home/>}
      />

      <Route
        path="/trainers"
        element={<TrainersList/>}
      />

      <Route
        path="/trainer/:id"
        element={<TrainerDetails/>}
      />

    </Routes>

  </BrowserRouter>

);

}

export default App;
```

Lab 7 - Props

Objective

Display Shopping Cart using Props.

Folder Structure

```
src/
├── Cart.js
├── OnlineShopping.js
└── App.js
```

Cart.js

```
function Cart(props){  
  return(  
    <table border="1">  
      <thead>  
        <tr>  
          <th>Item</th>  
          <th>Price</th>  
        </tr>  
      </thead>  
      <tbody>  
        {props.items.map(item=>(  
          <tr key={item.itemname}>  
            <td>{item.itemname}</td>  
            <td>{item.price}</td>  
          </tr>  
        ))}  
      </tbody>  
    </table>  
  );  
}  
  
export default Cart;
```

OnlineShopping.js

```
import Cart from "./Cart";  
  
function OnlineShopping(){  
  const items=[  
    {itemname:"Laptop",price:60000},
```

```
    {itemname:"Mouse",price:800},
    {itemname:"Keyboard",price:1500},
    {itemname:"Monitor",price:12000},
    {itemname:"Headphones",price:2500}

  ];

  return(

    <div>

      <h2>Shopping Cart</h2>

      <Cart items={items}/>

    </div>

  );

}

export default OnlineShopping;
```

App.js

```
import OnlineShopping from "./OnlineShopping";

function App(){

  return <OnlineShopping/>;

}

export default App;
```

Lab 8 - React State

Objective

Mall Entry Counter using State.

Folder Structure

```
src/
|— CountPeople.js
```

└─ App.js

CountPeople.js

```
import React,{Component} from "react";

class CountPeople extends Component{

  constructor(){

    super();

    this.state={

      entrycount:0,
      exitcount:0

    };

  }

  updateEntry={()=>{

    this.setState({

      entrycount:this.state.entrycount+1

    });

  }};

  updateExit={()=>{

    this.setState({

      exitcount:this.state.exitcount+1

    });

  }};

  render(){

    return(

      <div>

        <h2>Entry Count : {this.state.entrycount}</h2>

        <h2>Exit Count : {this.state.exitcount}</h2>

      </div>

    );

  }

}
```

```
        <button onClick={this.updateEntry}>
          Login
        </button>

        <button onClick={this.updateExit}>
          Exit
        </button>

      </div>

    );
  }
}

export default CountPeople;
```

App.js

```
import CountPeople from "./CountPeople";

function App(){
  return <CountPeople/>;
}

export default App;
```

React HOL Solutions - Part 3

Covers Labs 9–12

Lab 9 – ES6 Features

Objective

Implement:

- map()
- filter()
- Arrow Functions
- Destructuring
- Spread Operator

Folder Structure

```
src/  
├── ListofPlayers.js  
├── IndianPlayers.js  
└── App.js
```

ListofPlayers.js

```
function ListofPlayers(){  
  
  const players=[  
    {name:"Virat",score:95},  
    {name:"Rohit",score:82},  
    {name:"Gill",score:55},  
    {name:"Rahul",score:68},  
    {name:"Hardik",score:88},  
    {name:"Jadeja",score:71},  
    {name:"Surya",score:99},  
    {name:"Pant",score:66},  
    {name:"Ashwin",score:74},  
    {name:"Shami",score:48},  
    {name:"Bumrah",score:91}  
  ];  
  
  const below70=players.filter(  
    player=>player.score<70  
  );  
  
  return(  
  
    <div>  
  
      <h2>All Players</h2>  
  
      <ul>  
  
        {players.map(player=>(  
  
          <li key={player.name}>  
            {player.name} - {player.score}  
          </li>  
  
        ))}  
  
      </ul>  
  
      <h2>Players below 70</h2>
```

```
        <ul>

        {below70.map(player=>(

            <li key={player.name}>
                {player.name}
            </li>

        )))}

        </ul>

    </div>

    );

}

export default ListofPlayers;
```

IndianPlayers.js

```
function IndianPlayers(){

    const players=[
        "Virat",
        "Rohit",
        "Gill",
        "Rahul",
        "Hardik",
        "Jadeja"
    ];

    const [
        odd1,
        even1,
        odd2,
        even2,
        odd3,
        even3
    ]=players;

    const t20=[
        "Virat",
        "Rohit",
        "Gill"
    ];

    const ranji=[
        "Pujara",
```

```
    "Rahane",
    "Iyer"
  ];

  const merged=[
    ...t20,
    ...ranji
  ];

  return(

    <div>

      <h2>Odd Players</h2>

      <p>{odd1}</p>
      <p>{odd2}</p>
      <p>{odd3}</p>

      <h2>Even Players</h2>

      <p>{even1}</p>
      <p>{even2}</p>
      <p>{even3}</p>

      <h2>Merged Team</h2>

      <ul>

        {merged.map(player=>(

          <li key={player}>
            {player}
          </li>

        )))}

      </ul>

    </div>

  );

}

export default IndianPlayers;
```

App.js

```
import ListofPlayers from "./ListofPlayers";
import IndianPlayers from "./IndianPlayers";

function App(){

  const flag=true;

  return(

    <div>

      {flag
      ?
      <ListofPlayers/>
      :
      <IndianPlayers/>
      }

    </div>

  );

}

export default App;
```

Lab 10 – JSX

Objective

Display Office Space details using JSX and inline CSS.

Folder Structure

```
src/
└─ App.js
```

App.js

```
function App(){

  const office={
```

```
    name: "Tech Park",

    rent: 70000,

    address: "Bhopal"

  };

  const offices = [

    {
      name: "ABC Tower",
      rent: 55000,
      address: "Delhi"
    },

    {
      name: "XYZ Plaza",
      rent: 85000,
      address: "Mumbai"
    }

  ];

  return (

    <div>

      <h1>Office Space Rental App</h1>

      

      <h2>{office.name}</h2>

      <p
        style={{
          color: office.rent < 60000
            ?
            "red"
            :
            "green"
        }}
      >
        Rent : {office.rent}
      </p>

      <p>{office.address}</p>

      <hr/>

      {offices.map((item, index) => (
```

```
        <div key={index}>

            <h3>{item.name}</h3>

            <p
              style={{
                color:item.rent<60000
                ?
                "red"
                :
                "green"
              }}
            >
              Rent : {item.rent}
            </p>

            <p>{item.address}</p>

        </div>

    )))

</div>

);

}

export default App;
```

Lab 11 – Event Handling

Objective

Implement different Event Handlers.

Folder Structure

```
src/
├── Counter.js
├── CurrencyConvertor.js
└── App.js
```

Counter.js

```
import {Component} from "react";

class Counter extends Component{

  state={count:0};

  increment=()=>{

    this.setState({
      count:this.state.count+1
    });

  };

  decrement=()=>{

    this.setState({
      count:this.state.count-1
    });

  };

  sayHello=()=>{

    alert("Hello User");

  };

  welcome=(msg)=>{

    alert(msg);

  };

  clicked=()=>{

    alert("I was clicked");

  };

  increase=()=>{

    this.increment();

    this.sayHello();

  };

  render(){

    return(
```

```
        <div>

            <h2>{this.state.count}</h2>

            <button onClick={this.increase}>
                Increment
            </button>

            <button onClick={this.decrement}>
                Decrement
            </button>

            <button
                onClick={()=>this.welcome("Welcome")}
            >
                Say Welcome
            </button>

            <button onClick={this.clicked}>
                OnPress
            </button>

        </div>

    );
}

}

export default Counter;
```

CurrencyConvertor.js

```
import {useState} from "react";

function CurrencyConvertor(){

    const [rupees,setRupees]=useState("");

    const convert=()=>{

        const euro=(rupees/90).toFixed(2);

        alert(euro+" Euro");

    };

    return(
```



```
        <div>

            <input

                type="number"

                onChange={
                    e=>setRupees(e.target.value)
                }

            />

            <button onClick={convert}>
                Convert
            </button>

        </div>

    );
}

export default CurrencyConvertor;
```

App.js

```
import Counter from "../Counter";
import CurrencyConvertor from "../CurrencyConvertor";

function App(){

    return(

        <div>

            <Counter/>

            <hr/>

            <CurrencyConvertor/>

        </div>

    );

}

export default App;
```

Lab 12 – Conditional Rendering

Objective

Display Guest/User pages depending on Login status.

Folder Structure

```
src/  
├─ Guest.js  
├─ User.js  
└─ App.js
```

Guest.js

```
function Guest(){  
  
  return(  
  
    <div>  
  
      <h2>Guest Page</h2>  
  
      <p>Browse Flights</p>  
  
    </div>  
  
  );  
  
}  
  
export default Guest;
```

User.js

```
function User(){  
  
  return(  
  
    <div>  
  
      <h2>User Page</h2>
```

```
        <button>
          Book Ticket
        </button>

      </div>

    );
  }

  export default User;
```

App.js

```
import {useState} from "react";
import Guest from "../Guest";
import User from "../User";

function App(){

  const [login,setLogin]=useState(false);

  return(

    <div>

      <button
        onClick={()=>
          setLogin(!login)
        }
      >
        {login ? "Logout":"Login"}
      </button>

      <hr/>

      {
        login
        ?
        <User/>
        :
        <Guest/>
      }

    </div>

  );
}
```

```
export default App;
```

React HOL Solutions - Part 4

Covers Labs 13–16

Lab 13 – Conditional Rendering, Lists & Keys

Objective

Create a Blogger App with:

- Book Details
- Blog Details
- Course Details

using Conditional Rendering and Lists.

Folder Structure

```
src/  
├─ BookDetails.js  
├─ BlogDetails.js  
├─ CourseDetails.js  
└─ App.js
```

BookDetails.js

```
function BookDetails(){  
  
  const books=[  
  
    {  
      id:1,  
      name:"Java",  
      price:500  
    },  
  
    {  
      id:2,  
      name:"React",  
      price:650  
    }  
  ]  
}
```

```
    }

    ];

    return(

      <div>

        <h2>Book Details</h2>

        <ul>

          {books.map(book=>(

            <li key={book.id}>

              {book.name}

              - ₹{book.price}

            </li>

          ))}

        </ul>

      </div>

    );

  }

  export default BookDetails;
```

BlogDetails.js

```
function BlogDetails(){

  const blogs=[

    {

      id:1,

      title:"React Hooks"

    },

    {

      id:2,

      title:"NodeJS"

    }

  ]

}
```

```
];

return(

  <div>

    <h2>Blog Details</h2>

    <ul>

      {blogs.map(blog=>(

        <li key={blog.id}>

          {blog.title}

        </li>

      ))}

    </ul>

  </div>

);

}

export default BlogDetails;
```

CourseDetails.js

```
function CourseDetails(){

  const courses=[

    {

      id:1,

      name:"React"

    },

    {

      id:2,

      name:"Angular"

    },

    {

      id:3,

      name:"NodeJS"

    }

  ]

}
```

```
];

return(

  <div>

    <h2>Course Details</h2>

    <ul>

      {courses.map(course=>(

        <li key={course.id}>

          {course.name}

        </li>

      ))}

    </ul>

  </div>

);

}

export default CourseDetails;
```

App.js

```
import BookDetails from "./BookDetails";
import BlogDetails from "./BlogDetails";
import CourseDetails from "./CourseDetails";

function App(){

  const choice="book";

  if(choice==="book")
    return <BookDetails/>;

  if(choice==="blog")
    return <BlogDetails/>;

  return <CourseDetails/>;

}
```

```
export default App;
```

Lab 14 – React Context API

Objective

Replace Props with Context API.

Folder Structure

```
src/  
├─ ThemeContext.js  
├─ EmployeeCard.js  
├─ EmployeeList.js  
└─ App.js
```

ThemeContext.js

```
import {createContext} from "react";  
  
const ThemeContext=createContext("light");  
  
export default ThemeContext;
```

EmployeeCard.js

```
import {useContext} from "react";  
import ThemeContext from "../ThemeContext";  
  
function EmployeeCard({employee}){  
  
  const theme=useContext(ThemeContext);  
  
  return(  
  
    <div  
      style={{  
        backgroundColor:  
          theme==="dark"  
        ?
```



```
        "#333"  
        :  
        "#eee",  
  
        color:  
        theme=== "dark"  
        ?  
        "white"  
        :  
        "black",  
  
        padding: "15px",  
  
        margin: "10px"  
    }}  
    >  
  
    <h3>{employee.name}</h3>  
  
    <p>{employee.designation}</p>  
  
    </div>  
  
    );  
  
}  
  
export default EmployeeCard;
```

EmployeeList.js

```
import EmployeeCard from "../EmployeeCard";  
  
function EmployeeList(){  
  
    const employees=[  
  
        {  
            id:1,  
            name:"Varun",  
            designation:"Developer"  
        },  
  
        {  
            id:2,  
            name:"Rahul",  
            designation:"Tester"  
        }  
  
    ];  
  
};
```

```
    return(  
      <div>  
        {employees.map(emp=>(  
          <EmployeeCard  
            key={emp.id}  
            employee={emp}  
          />  
        ))}  
      </div>  
    );  
  }  
  
  export default EmployeeList;
```

App.js

```
import {useState} from "react";  
  
import ThemeContext from "../ThemeContext";  
  
import EmployeeList from "../EmployeeList";  
  
function App(){  
  const [theme,setTheme]=useState("light");  
  
  return(  
    <ThemeContext.Provider  
      value={theme}  
    >  
  
      <button  
        onClick={()=>  
          setTheme(  
            theme==="light"  
            ?  
            "dark"          )  
        }  
      />  
    </ThemeContext.Provider>  
  )  
}
```

```
        :  
        "light"  
      )  
    }  
  >  
  
    Change Theme  
  
  </button>  
  
  <EmployeeList/>  
  
</ThemeContext.Provider>  
  
);  
  
}  
  
export default App;
```

Lab 15 – React Forms

Objective

Complaint Registration Form

Folder Structure

```
src/  
├── ComplaintRegister.js  
└── App.js
```

ComplaintRegister.js

```
import {useState} from "react";  
  
function ComplaintRegister(){  
  
  const [name, setName]=useState("");  
  
  const [complaint, setComplaint]=useState("");  
  
  const handleSubmit=(e)=>{
```

```
e.preventDefault();

const ref=Math.floor(

    Math.random()*100000

);

alert(

    "Complaint Registered\nReference No : "

    +ref

);

};

return(

    <form onSubmit={handleSubmit}>

        <h2>

            Raise Complaint

        </h2>

        <input

            placeholder="Employee Name"

            value={name}

            onChange={

                e=>setName(e.target.value)

            }

        />

        <br/><br/>

        <textarea

            placeholder="Complaint"

            value={complaint}

            onChange={

                e=>setComplaint(e.target.value)
```

```
        }  
      />  
  
      <br/><br/>  
  
      <button>  
        Submit  
      </button>  
  
    </form>  
  );  
}  
  
export default ComplaintRegister;
```

App.js

```
import ComplaintRegister from "./ComplaintRegister";  
  
function App(){  
  return <ComplaintRegister/>;  
}  
  
export default App;
```

Lab 16 – Form Validation

Objective

Validate:

- Name $\geq$ 5 characters
- Email contains @ and .
- Password $\geq$ 8 characters

Folder Structure

```
src/  
├─ Register.js  
└─ App.js
```

Register.js

```
import {useState} from "react";  
  
function Register(){  
  
  const [name,setName]=useState("");  
  
  const [email,setEmail]=useState("");  
  
  const [password,setPassword]=useState("");  
  
  const submit=(e)=>{  
  
    e.preventDefault();  
  
    if(name.length<5){  
  
      alert("Invalid Name");  
  
      return;  
  
    }  
  
    if(  
  
      !email.includes("@")  
  
      ||  
  
      !email.includes(".")  
  
    ){  
  
      alert("Invalid Email");  
  
      return;  
  
    }  
  
    if(password.length<8){  
  
      alert("Invalid Password");  
  
      return;  
  
    }  
  
  }  
  
}
```

```
    }

    alert("Registration Successful");

  };

  return(

    <form onSubmit={submit}>

      <input

        placeholder="Name"

        value={name}

        onChange={

          e=>setName(e.target.value)

        }

      />

      <br/><br/>

      <input

        placeholder="Email"

        value={email}

        onChange={

          e=>setEmail(e.target.value)

        }

      />

      <br/><br/>

      <input

        type="password"

        placeholder="Password"

        value={password}

        onChange={

          e=>setPassword(e.target.value)
```

```
    }  
  />  
  
  <br/><br/>  
  
  <button>  
    Register  
  </button>  
  
</form>  
);  
}  
  
export default Register;
```

App.js

```
import Register from "./Register";  
  
function App(){  
  return <Register/>;  
}  
  
export default App;
```

React HOL Solutions - Part 5

Covers Labs 17–19

Lab 17 – Fetch API (REST API)

Objective

Fetch user details from **<https://api.randomuser.me/>** and display:

- Title
- First Name

- Image

Folder Structure

```
src/  
├─ GetUser.js  
└─ App.js
```

GetUser.js

```
import React,{Component} from "react";  
  
class GetUser extends Component{  
  
  constructor(){  
  
    super();  
  
    this.state={  
  
      user:null  
  
    };  
  
  }  
  
  async componentDidMount(){  
  
    const response=await fetch(  
      "https://randomuser.me/api/"  
    );  
  
    const data=await response.json();  
  
    this.setState({  
  
      user:data.results[0]  
  
    });  
  
  }  
  
  render(){  
  
    if(this.state.user===null){  
  
      return <h2>Loading...</h2>;  
  
    }  
  
  }  
}
```

```
    }

    const user=this.state.user;

    return(

      <div>

        <img
          src={user.picture.large}
          alt="User"
        />

        <h2>

          {user.name.title}

          {" "}

          {user.name.first}

        </h2>

      </div>

    );

  }

}

export default GetUser;
```

App.js

```
import GetUser from "../GetUser";

function App(){

  return <GetUser/>;

}

export default App;
```

Lab 18 – Unit Testing using Jest & Enzyme

Objective

Write Unit Tests for CohortDetails component.

Install

```
npm install enzyme
npm install enzyme-adapter-react-16
```

setupTests.js

```
import Enzyme from "enzyme";
import Adapter from "enzyme-adapter-react-16";

Enzyme.configure({
  adapter: new Adapter()
});
```

CohortDetails.test.js

```
import {shallow,mount} from "enzyme";

import CohortDetails from "../CohortDetails";

describe(
  "Cohort Details Component",
  ()=>{
    const cohort={
      code:"INTADMDF10",
      trainer:"John",
      strength:25,
      status:"ongoing"
    };
  });
```

```
test(
  "should create component",
  ()=>{
    const wrapper=shallow(
      <CohortDetails cohort={cohort}/>
    );
    expect(wrapper.exists()).toBe(true);
  }
);

test(
  "should initialize props",
  ()=>{
    const wrapper=mount(
      <CohortDetails cohort={cohort}/>
    );
    expect(
      wrapper.props().cohort
    ).toEqual(cohort);
  }
);

test(
  "should display cohort code",
  ()=>{
    const wrapper=mount(
      <CohortDetails cohort={cohort}/>
    );
    expect(
```

```
wrapper.find("h3").text()

).toBe("INTADMDF10");

}

);

test(

"should match snapshot",

()=>{

const wrapper=shallow(

<CohortDetails cohort={cohort}/>

);

expect(

wrapper

).toMatchSnapshot();

}

);

});
```

Run Tests

```
npm test
```

Lab 19 – Mocking Axios using Jest

Objective

Fetch GitHub repositories and Mock Axios.

Install

```
npm install axios
```

GitClient.js

```
import axios from "axios";

class GitClient{

  async getRepositories(user){

    const response=await axios.get(

      `https://api.github.com/users/${user}/repos`

    );

    return response.data;

  }

}

export default new GitClient();
```

App.js

```
import {useEffect,useState} from "react";

import GitClient from "./GitClient";

function App(){

  const [repos,setRepos]=useState([]);

  useEffect(()=>{

    GitClient

    .getRepositories("techiesyed")

    .then(data=>setRepos(data));

  },[]);

  return(
```

```
        <div>

            <h2>

                GitHub Repositories

            </h2>

            <ul>

                {repos.map(repo=>(

                    <li key={repo.id}>

                        {repo.name}

                    </li>

                ))}

            </ul>

        </div>

    );

}

export default App;
```

GitClient.test.js

```
import axios from "axios";

import GitClient from "../GitClient";

jest.mock("axios");

describe(

    "Git Client Tests",

    ()=>{

        test(

            "should return repository names for techiesyed",

            async()=>{
```

```
const repos=[

  {
    id:1,
    name:"ReactProject"
  },

  {
    id:2,
    name:"NodeProject"
  }

];

axios.get.mockResolvedValue({

  data:repos

});

const response=

await GitClient.getRepositories(

"techiesyed"

);

expect(

response.length

).toBe(2);

expect(

response[0].name

).toBe("ReactProject");

});

});
```

Run Tests

```
npm test
```